



Network Operations & GSM at COMM4 Telecom Pvt. Ltd.

SUMMER INTERNSHIP PRESENTATION

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About COMM4 Telecom Pvt. Ltd.

- ▶ **COMM4 Telecom Pvt. Ltd.** is a telecom infrastructure and networking solutions company, operating out of iThum Tower, Sector-62, Noida, with a registered office in Delhi.

The company works in the telecom domain, covering network infrastructure and operations that support mobile and telecom connectivity.

INDUSTRY

Telecom Infrastructure

LOCATION

Sector-62, Noida (Corp. Office)

INTERNSHIP TRACK

Network Operations

Objective of the Internship

1

Industry Exposure

Understand how a real telecom company runs its network operations, beyond classroom theory.

2

Core Networking Concepts

Build a working understanding of Cisco networking fundamentals used in enterprise/telecom setups.

3

Telecom Domain Knowledge

Learn the basics of GSM network architecture and how mobile networks function end-to-end.

4

Professional Exposure

Observe day-to-day operations, documentation practices, and team workflows in a live NOC-style environment.

Internship Domain: Network Operations

The internship was based in the Network Operations vertical, which sits at the intersection of enterprise networking and telecom (GSM) infrastructure.

Enterprise Networking

- Cisco routers, switches & IOS concepts
- IP addressing & subnetting basics
- Routing vs. switching fundamentals
- Network topology & documentation

Telecom (GSM) Infrastructure

- GSM network architecture (BTS, BSC, MSC)
- Basics of mobile call flow
- Role of network operations in uptime
- How enterprise & telecom networks interlink

Cisco Networking — Concepts Learned

This was primarily a theoretical / conceptual learning track — building a foundation in Cisco networking without live device configuration.

- 1 Networking Devices** Understood the role of routers and switches in an enterprise network, and how Cisco devices fit into that setup.
- 2 IP Addressing** Learned IPv4 addressing, subnetting basics, and how devices are logically organized on a network.
- 3 Routing & Switching** Studied the conceptual difference between routing (between networks) and switching (within a network).
- 4 Cisco IOS Basics** Got introduced to the Cisco IOS command-line environment and common configuration terminology.

GSM Network Fundamentals

Exposure to how a mobile (GSM) network is structured, from the handset to the core network.



BTS (Base Transceiver Station): handles radio communication with mobile handsets.

BSC (Base Station Controller): manages multiple BTS units and handles handovers.

MSC (Mobile Switching Center): routes calls and connects the network to other telecom systems

Day-to-Day Activities

1

Learning Sessions

Attended structured sessions on networking and GSM concepts conducted by the team.

2

Hands-on Tasks

Worked on select hands-on exercises to reinforce concepts covered in learning sessions.

3

Shadowing

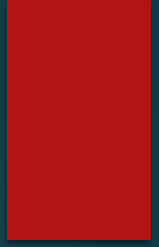
Observed engineers handling network monitoring and operational discussions.

4

Documentation Support

Assisted with organizing notes and reference material on networking topics covered.

Tools & Technologies



1

Cisco IOS / Concepts

Command-line fundamentals and device configuration terminology.

2

GSM Architecture Models

Reference diagrams and material explaining mobile network structure.

3

MS Office (Excel/Word)

Used for notes, documentation, and organizing learning material.

4

Networking Reference Material

Company-provided training content on telecom network operations.

Key Learnings & Skills Gained



Conceptual grounding in Cisco networking fundamentals (routing, switching, IP addressing).



Understanding of GSM network architecture and how mobile networks operate.



Exposure to how a real telecom company structures its network operations.



Improved technical documentation and note-organization habits.



Better understanding of professional workplace communication and teamwork.

Challenges Faced



New Terminology

Telecom and GSM-specific terms were unfamiliar at first and took time to absorb.



Theory Without Live Practice

Learning Cisco concepts without hands-on device access made some ideas harder to fully internalize.



Short Duration

A one-month window limited how deep the exposure to each topic could go.

Conclusion

This internship at COMM4 Telecom Pvt. Ltd. provided a valuable first look into the telecom networking industry — building a conceptual foundation in Cisco networking and GSM architecture, and offering exposure to how network operations function in a real organization.

It has laid a strong groundwork for further, deeper learning in networking and telecom systems.

Thank You

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